ECON 104 Module 2: The Story

Lectures 8–16 · from growth to business cycles · Samrudha Surana

The through-line. Why are some countries rich, and why does the economy fluctuate around its growth path? Growth comes from capital and good institutions; to track it we measure output (GDP), labor (unemployment) and prices (inflation); and AD–AS explains why real GDP slips below potential.

1. Economic growth

- **Economic growth is enough, and only economic growth is enough.** Everyone used to be poor: for most of history people lived much as their ancestors had, and sustained growth only began in the 19th century. Today incomes and living standards are enormously higher (electricity, clean water, falling child mortality, longer lives), yet income per person still varies hugely across countries.
- **Miracles and disasters.** Japan grew 8.5% a year for 20 years after WWII (GDP per person doubles every 8 years) and South Korea 7.2% from 1970 to 1990. Argentina fell from 75% of U.S. income per person in 1900 to 36% today. Growth is a PPF shifting out, which is the same thing as demand shifting out as wages rise.
- **Proximate causes: capital creation comes first.** The island story: consume less today (save), invest the time in a spear (capital), and consume more tomorrow. Capital makes labor more productive (a tractor vs. a hand plow, which is why U.S. farm workers earn more than Indian ones). It comes in three forms: physical capital, human capital (education, training, even internships) and technological knowledge (1,000 shovels vs. one machine). Specialization and trade add to it, but the division of labor is limited by the extent of the market.
- **Deeper causes: why do some countries build more capital?** Nobody builds the spear or the boat if it can be stolen, so secure, enforceable **property rights** (including against the state) come first. **Prices** carry information and incentives, and **profit and loss** reward people who create value for others and punish waste, which pushes resources toward their most productive uses and encourages investment, innovation and risk-taking. Together they make investments safer and more productive. The two Koreas share people, culture and geography, but per capita income in the South is about 28 times the North's ($36,600 vs. $1,300).

2. Saving and investment

- **Definitions.** Saving is deferring consumption today to consume more tomorrow; investment is the purchase, creation or maintenance of capital. People save to invest and to smooth consumption over a lifetime, and they borrow because they are impatient (time preference), to fund big projects, or to smooth consumption (college).
- **The interest rate is the price of loanable funds.** Saving supplies funds (a higher rate means more saved); borrowing demands them (a lower rate means more borrowed).
- **Equilibrium and shifts.** Above equilibrium there is a surplus of savings and lenders cut rates; below it there is a shortage and borrowers bid rates up.
  - Supply of funds up: rate falls, quantity of funds up. Supply down: rate rises, quantity down.
  - Demand for funds up: rate rises, quantity up. Demand down: rate falls, quantity down.
- **Banks, bonds and stocks.** Intermediaries cut the cost of connecting savers and borrowers. Banks coordinate time preferences and liquidity, judge projects, spread risk, and earn the spread between loan and deposit rates (more division of labor). A bond is an IOU, so the buyer is a lender. A stock is part ownership with a claim on future profits, which makes it riskier.
- **Bridge.** Growth is about standards of living, which is consumption. To consume we need to produce, to have incomes to trade, and to work. So let us look at the other macro variables, starting with how we measure output.

3. GDP

- **Macro = whole economies** ("there may be macroeconomic questions, but only microeconomic answers"). GDP is the market value of all finished goods and services produced within a country in a year. Real GDP per capita is a rough measure of the standard of living, though it misses underground and non-priced production, pollution, and health.
- **Real vs. nominal.** When we measure GDP growth, part of it is real output and part is inflation. Real GDP holds prices fixed: in 2025 nominal GDP grew 5.00% but real GDP only 2.12%, so the rest was prices.
- **Two sides of one pie.** Expenditure approach: Y = C + I + G + NX. Income approach: Y = employee compensation + rent + interest + profit. They match (about $30.76 trillion in 2025). The expenditure side is spending, which is what aggregate demand will describe.

4. Unemployment

- **Why we turn to labor.** To produce you need people employed, and labor is a key factor of production. In periods of low production unemployment is higher. How do we explain that?
- **The rate.** You are unemployed if you are 16+, not institutionalized, without a job and actively looking. Unemployment rate = unemployed / labor force (7.1M / 170.3M = 4.2% in September 2026). It is a rough indicator since it ignores job quality.
- **Three kinds.**
  - **Frictional** always exists and is usually healthy: matching workers and jobs takes time and information, and millions of jobs are created and destroyed every month.
  - **Structural** is persistent, long-term unemployment from skills mismatch, technological change, and regulation (generous benefits, minimum wages, hiring and firing costs). Long-standing structural unemployment is a problem; in France, Germany, Italy and Spain 40–50% of the unemployed have been out of work over a year.
  - **Cyclical** is the one that changes frequently, rising when real GDP growth is below average. It is often blamed on sticky wages (prices fall, wages do not). To understand it we need to think about economic fluctuations.
- **Natural rate = frictional + structural**, which changes slowly. Zero unemployment is not the goal.

5. Money and prices: inflation and the quantity theory

- **Inflation** is an increase in the average level of prices, measured with an index like the CPI. One price rising (eggs) is not inflation. Extreme cases show the stakes: Mugabe printed money to pay for his regime and Zimbabwe's inflation went from 50% a year to 50% a day.
- **The quantity theory: Mv = PY.** M is the money supply, v velocity (how often a dollar is spent), P the price level and Y real GDP. Both sides equal nominal GDP, so it is an identity. To make it a theory, assume real GDP is set by real factors (capital, labor, technology, as in section 1) and velocity is stable. Then money growth ≈ inflation. Friedman: "Inflation is always and everywhere a monetary phenomenon."
- **What is money?** A generally accepted medium of exchange (also a store of value and unit of account). Bank loans create it: $200 of currency becomes $270 of money after two rounds of lending. The Fed measures it as M1 and M2.
- **Neutral in the long run, not in the short run.** An unexpected increase in money can boost output at first; once firms and workers adjust, output grows no faster and only prices rise.
- **Costs of inflation:** price confusion and money illusion; redistribution (real rate = $i - \pi$, so surprise inflation helps borrowers, including the government, and hurts lenders); interactions with taxes (bracket creep, phantom profits); and it is painful to stop, since lower inflation can be misread as lower demand and cost jobs.

- **Bridge.** If money matters for output only in the short run, what happens in the short run? That brings us to business fluctuations.

6. Economic growth is not smooth: business cycles and AD–AS

- **The puzzle.** Real GDP averages about 3.1% growth, but there are large fluctuations: roughly −5% in a typical recession and 7–8% in a boom. Recessions idle labor and capital and push GDP below potential.
- **How do we explain this slack?** Say's law ("you cannot have demand without supply") says markets self-adjust and what is produced generates the income to buy it, so we should not see slack. Yet we do. Keynes challenged exactly this, and the AD–AS model is how we explain it.
- **Rewrite the quantity theory in growth rates: M + v = P + Y.** Spending growth (M + v) splits between inflation (P) and real growth (Y). With 5% spending growth, 0% real growth means 5% inflation; 3% real growth means 2%.
- **AD** shows all combinations of inflation and real growth consistent with a given rate of spending growth; it slopes down. It shifts right with faster money growth, confidence, wealth, lower taxes, faster government spending or export growth and slower import growth, and left with the opposites (2008: fear, AD left; COVID stimulus: AD right, more inflation).
- **LRAS** comes from the factors of supply (capital, labor and skills, technology), which fix potential growth. It is **vertical**: inflation of 0%, 2% or 200% does not change the number of factories. If spending grows 10% and LRAS is 3%, inflation is 7%. Real shocks shift it: right (good weather, cheaper oil, technology boom) raises growth and lowers inflation; left (oil, pandemic, war, higher taxes and regulation) lowers growth and raises inflation. The 1973 OPEC shock tripled oil prices in two years.
- **SRAS** slopes up because wages and prices are sticky: wages are fixed by contracts, firms face menu costs, workers suffer nominal wage confusion (mistaking nominal for real changes), and uncertainty. It shifts with **expected inflation**: higher expected inflation shifts it up, lower shifts it down.
- **The cycle.** An AD shock changes output in the short run (a → b). Then wages and expectations adjust, SRAS shifts, and growth returns to LRAS (c). In the long run AD moves only inflation. After a fall in AD, wages and prices are especially sticky downward, so the recession lingers before the adjustment. **COVID:** a negative real shock (LRAS left) plus a fall in AD pushed growth sharply negative; stimulus then moved AD back out and, as the model predicts, raised inflation.
- **So what can cause lower output?** (1) A negative AD shock. (2) A negative real shock moving LRAS left. (3) Higher inflation expectations shifting SRAS up. Which one it is tells us where to look.

Shift	Real GDP growth	Inflation
Increase in AD	↑ (short run)	↑
Decrease in AD	↓ (short run)	↓
Increase in SRAS	↑	↓
Decrease in SRAS	↓	↑
Increase in LRAS	↑	↓
Decrease in LRAS	↓	↑